



Introduction to HPC and Parallel Programming

Karuna Prasad
Scientist E
C-DAC, Bangalore
karunap@cdac.in

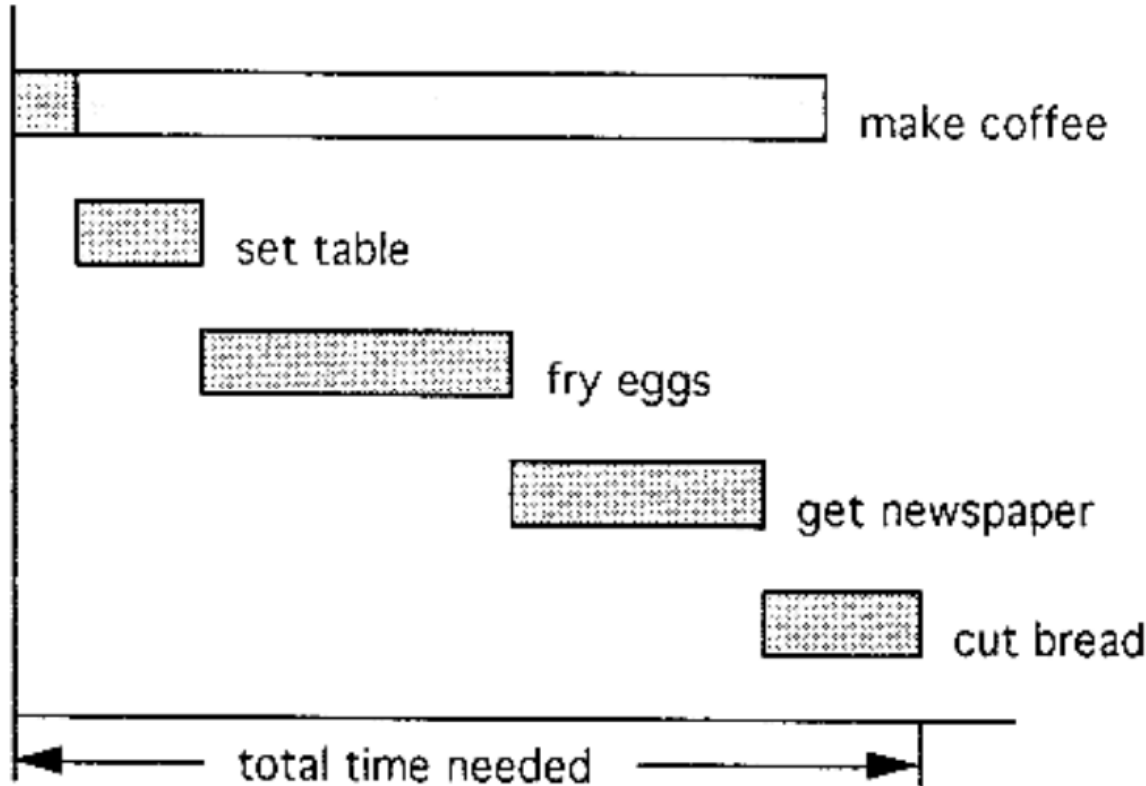


Agenda

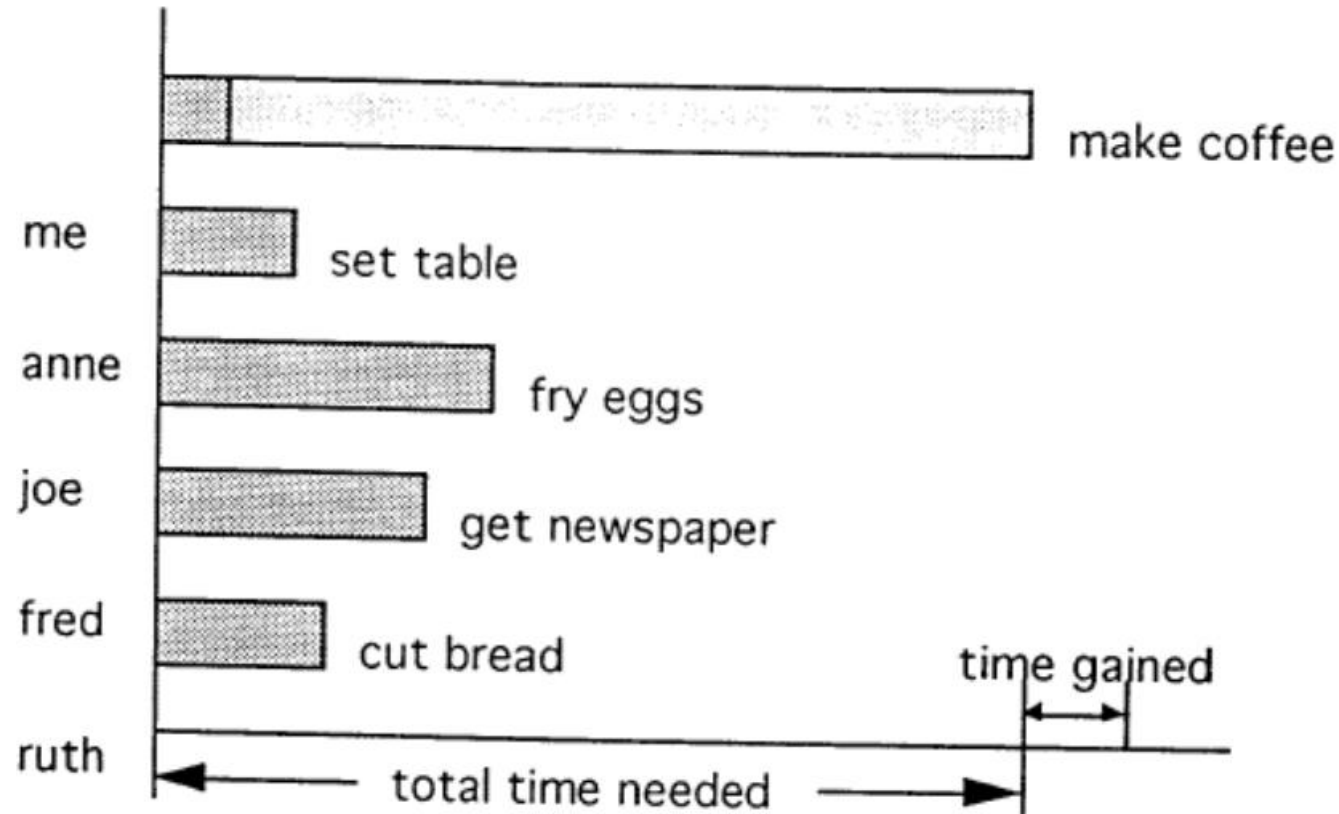


- Parallel Computing
- Parallel Architectures
- Memory Architectures
- Flynn's Taxonomy
- High Performance Computing in India
- NSM- Param Utkarsh Supercomputer
- Parallel Programming Models

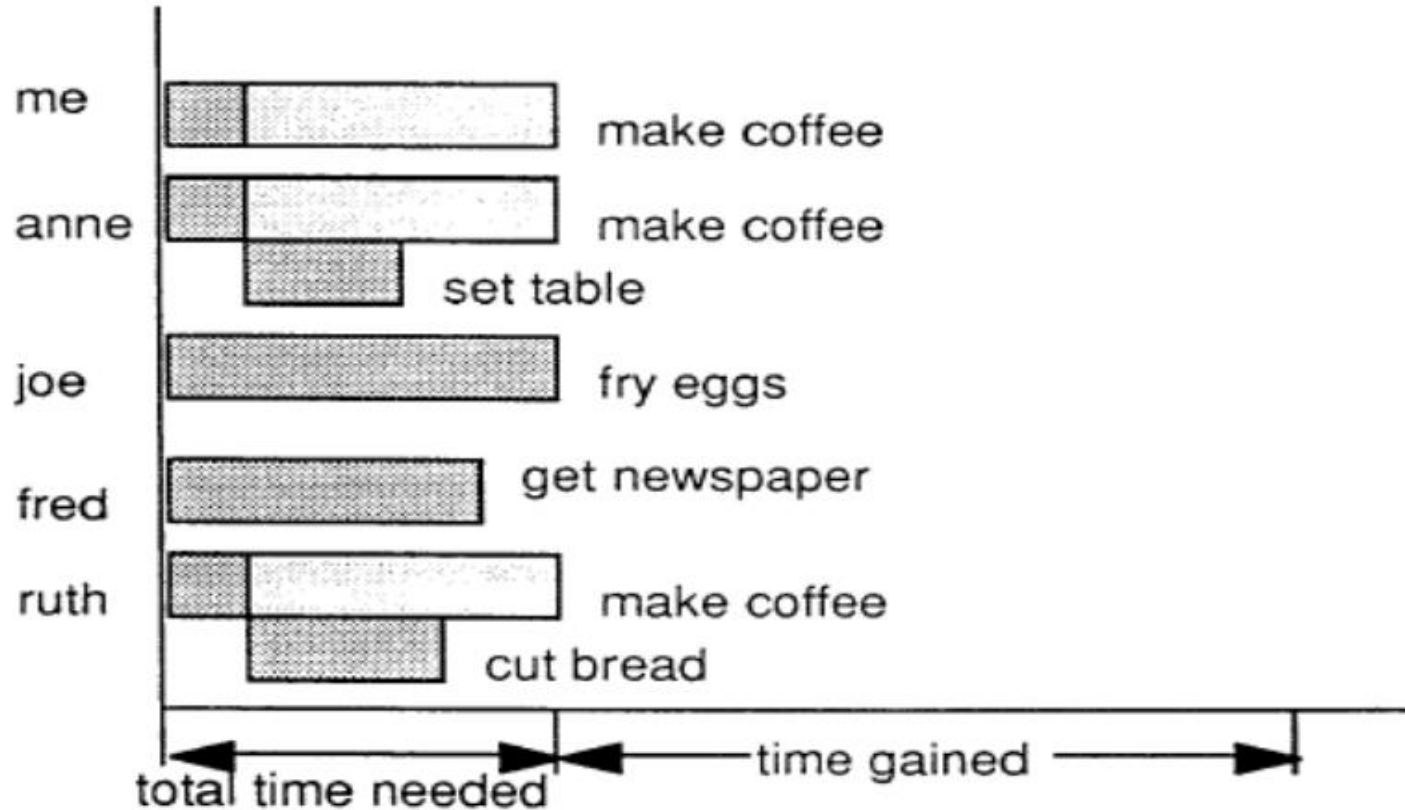
How I used to make breakfast....



How to set family to work



How finally got to the office in time....





Serial Computing



- If 1 person takes 15 hours to complete a job
- How long will it take if 15 person work together



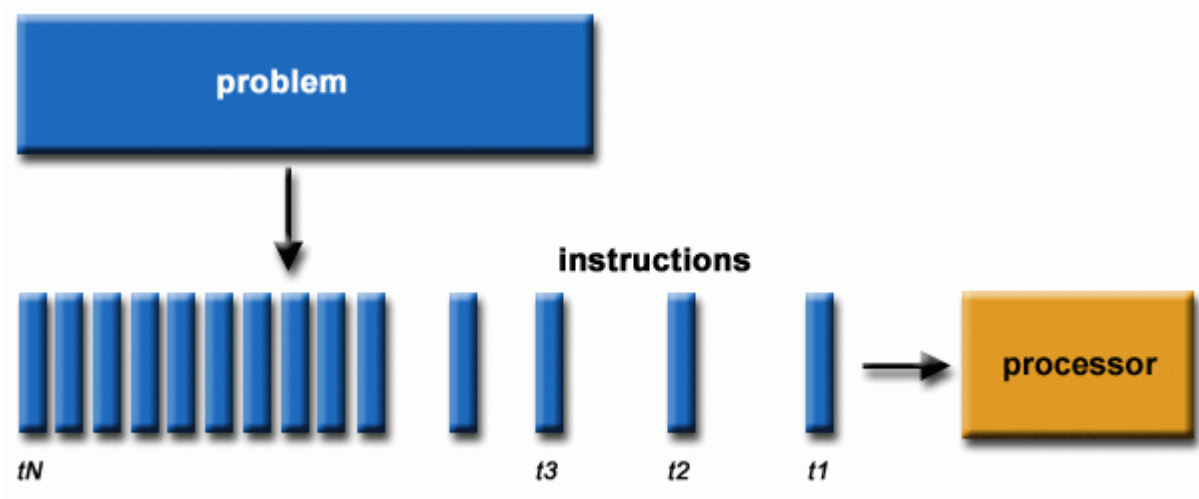
Parallel Computing

- Save time - wall clock time
- Solve larger problems
- Provide concurrency
- Make better use of underlying parallel hardware
- Take advantage of non-local resources

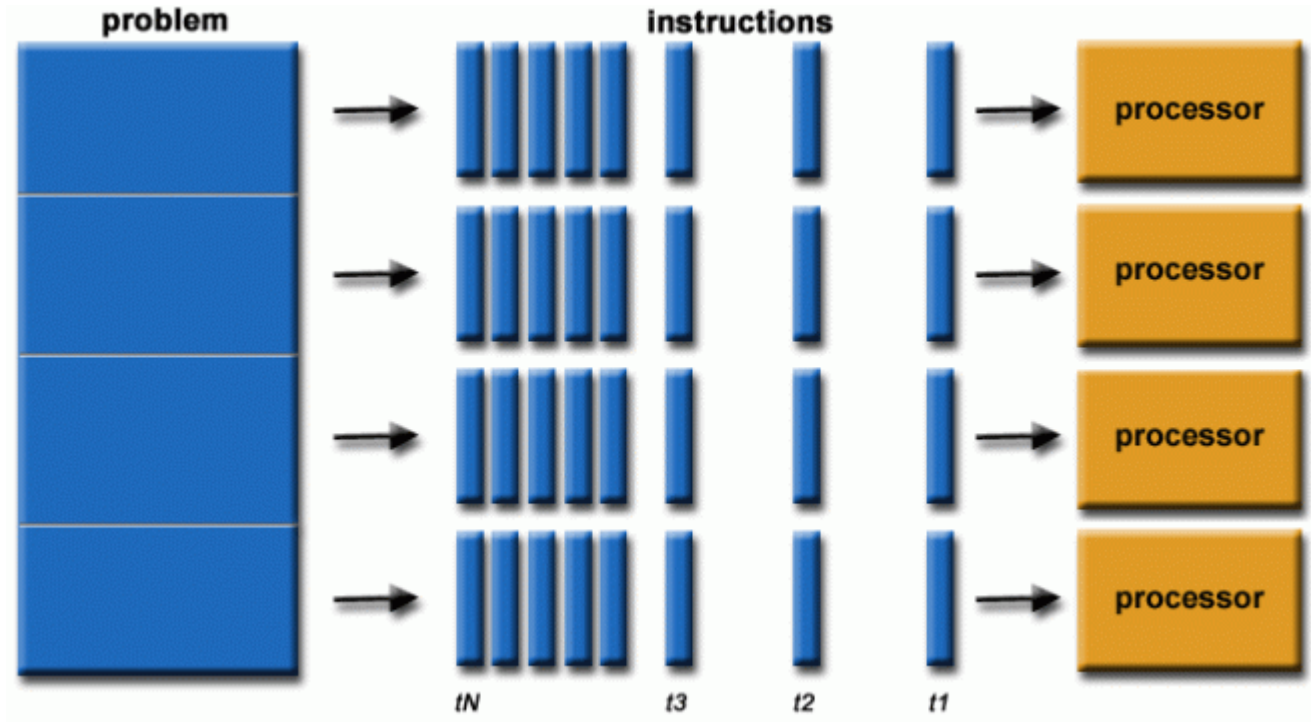
Parallelism is the necessity !!



Serial Computing



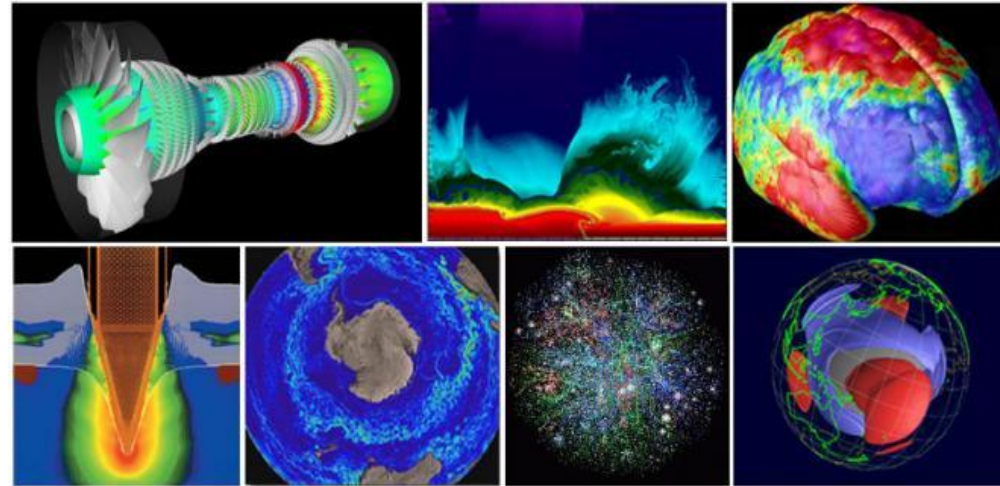
Parallel Computing



Uses of Parallel Computing

Science and Engineering

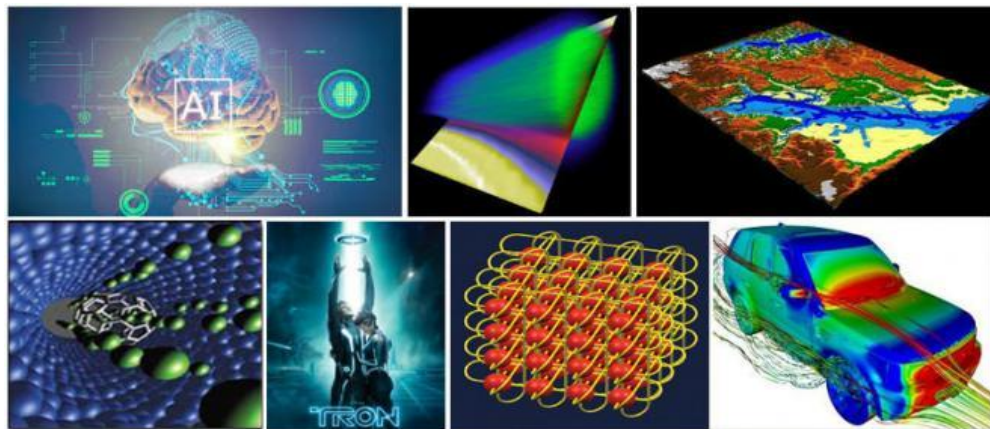
- Atmosphere, Earth, Environment
- Physics - applied, nuclear, particle, condensed matter, high pressure, fusion, photonics
- Bioscience, Biotechnology, Genetics
- Chemistry, Molecular Sciences
- Geology, Seismology
- Mechanical Engineering - from prosthetics to spacecraft
- Electrical Engineering, Circuit Design, Microelectronics
- Computer Science, Mathematics
- Defense, Weapons



Uses of Parallel Computing

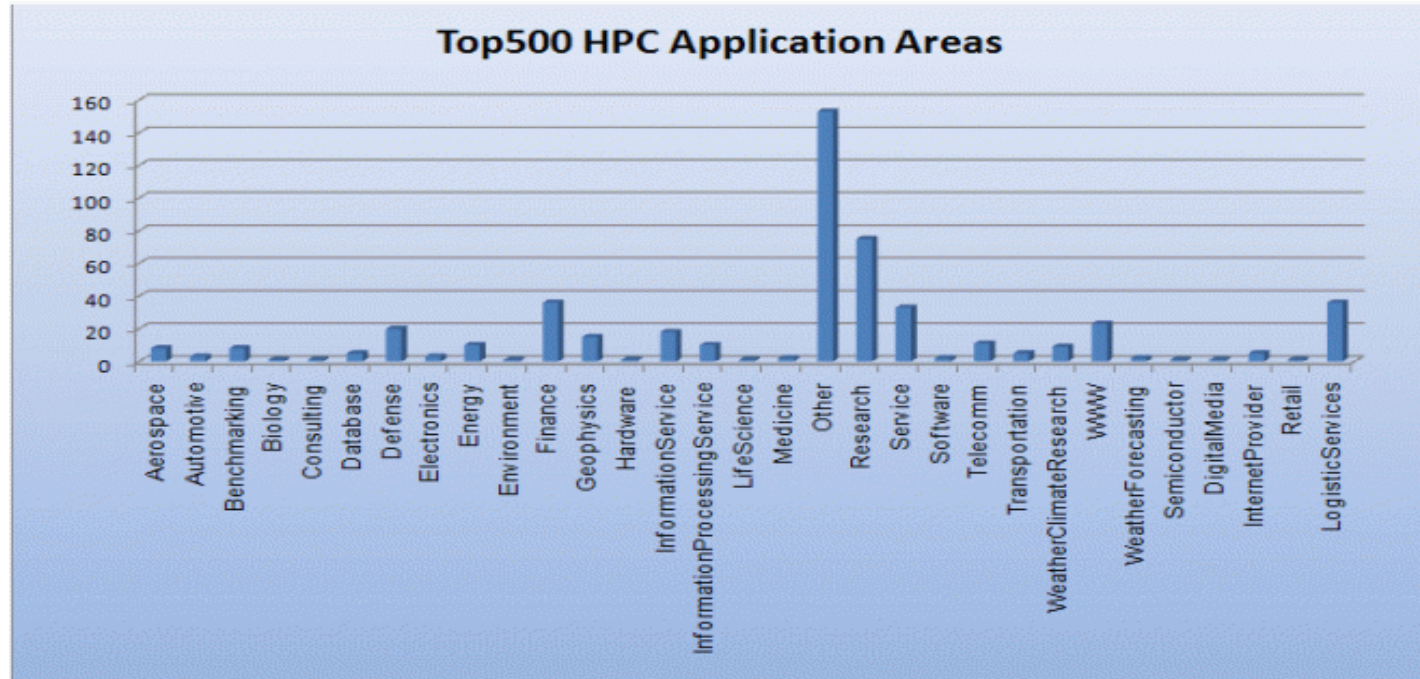
Industrial and Commercial

- Big Data, databases, data mining
- Artificial Intelligence (AI)
- Oil exploration
- Web search engines, web based business services
- Medical imaging and diagnosis
- Pharmaceutical design
- Financial and economic modeling
- Management of national and multi-national corporations
- Advanced graphics and virtual reality
- Networked video and multi-media technologies
- Collaborative work environments

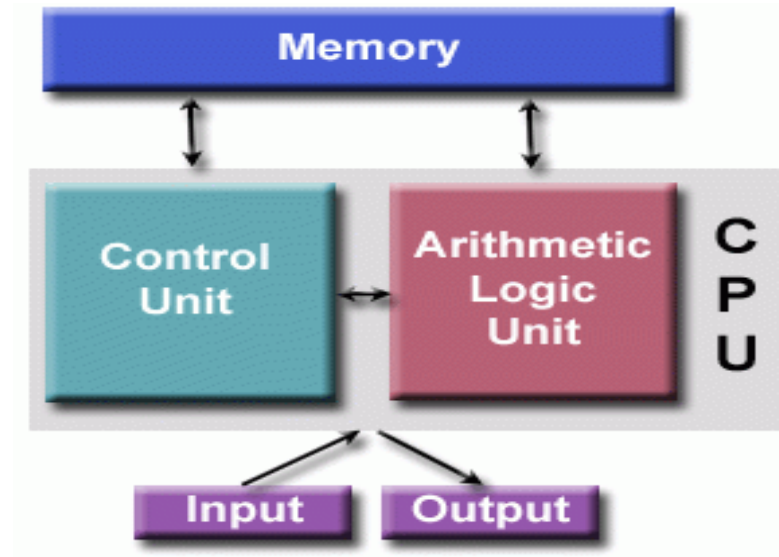


Uses of Parallel Computing

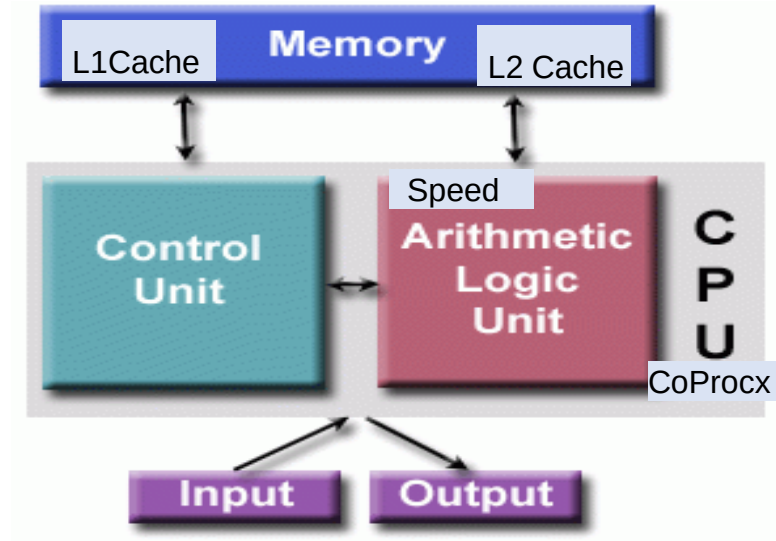
Global Applications



Simple Von Neumann Machine



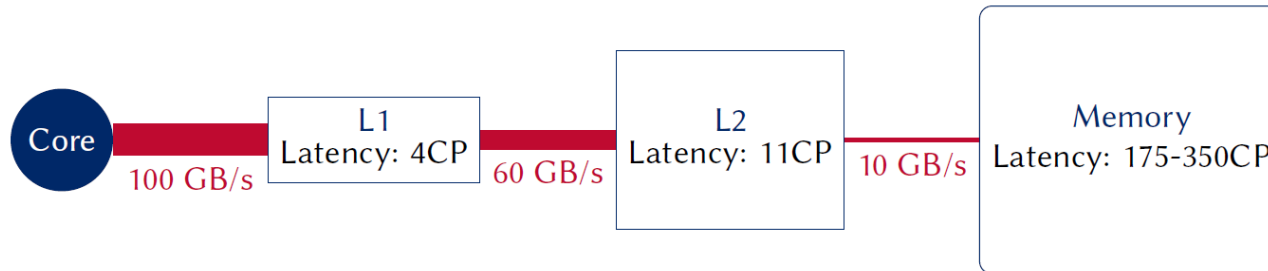
Von Neumann Architecture



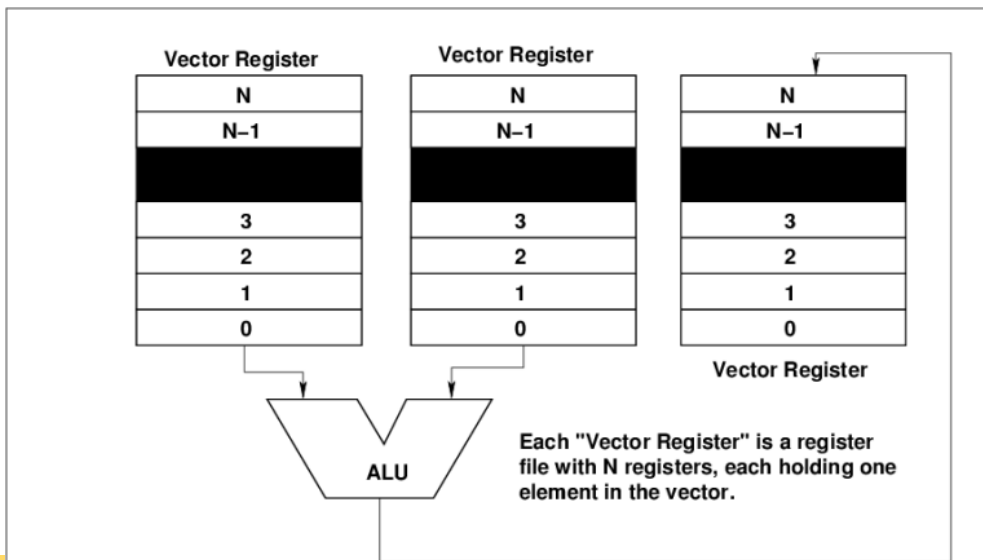
- Memory Hierarchy
 - Caches are smaller but faster (higher bandwidth, less latency) than memory
 - Multiple level of cache: L1, L2, L3
 - Helps keep cores fed with data
 - Cache reuse is critical to performance

Level	Size	Latency	Physical Location
L1 cache	32 KB	4 cycles	inside each core
L2 cache	256 KB	12 cycles	beside each core
L3 cache	6 MB	~21 cycles	shared between all cores
L4 E-cache	128 MB	~58 cycles	separate eDRAM chip
RAM	4+ GB	~117 cycles	SDRAM DIMMs on motherboard
Swap	100+ GB	10,000+ cycles	hard disk or SSD

Table 4 – The memory hierarchy of a modern desktop/laptop: Core i*4 Haswell.



- Vector Processors
 - able to run mathematical operations on multiple data elements simultaneously
 - Provides instruction that operate on entire array
 - Cray Machines



- Superscalar Processors
 - Instruction level parallelism with processor
 - Intel 960, SunSparc, MIPS R
 - and AMD Am29000

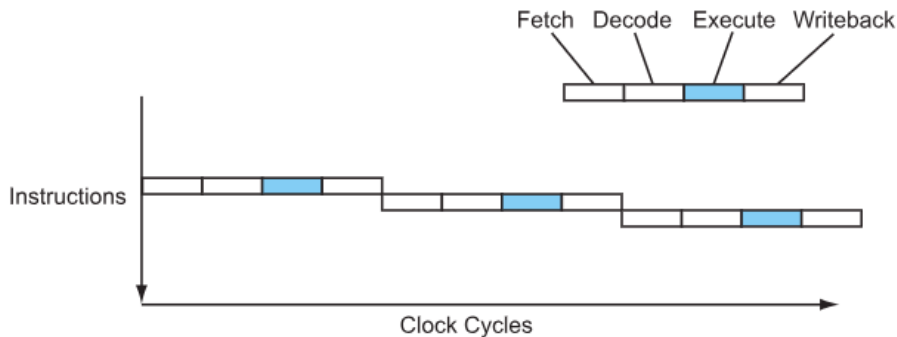


Figure 1 – The instruction flow of a sequential processor.

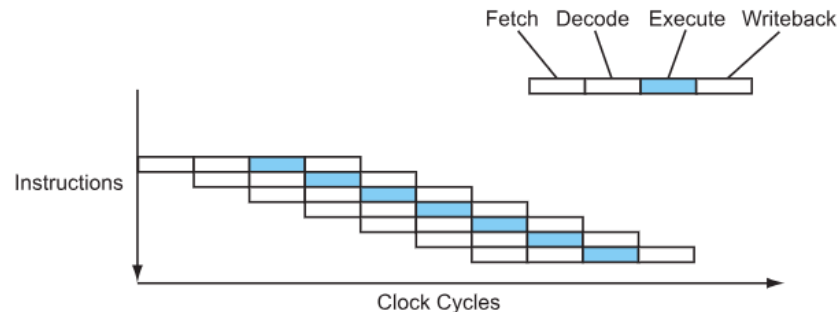
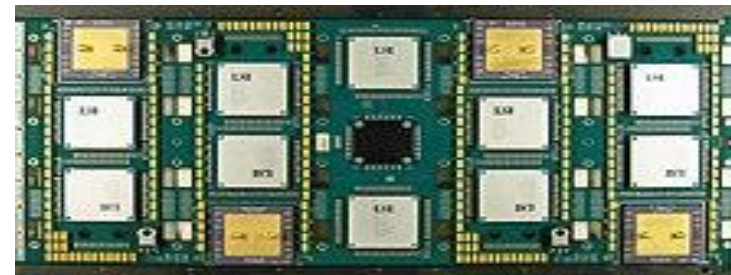


Figure 2 – The instruction flow of a pipelined processor.

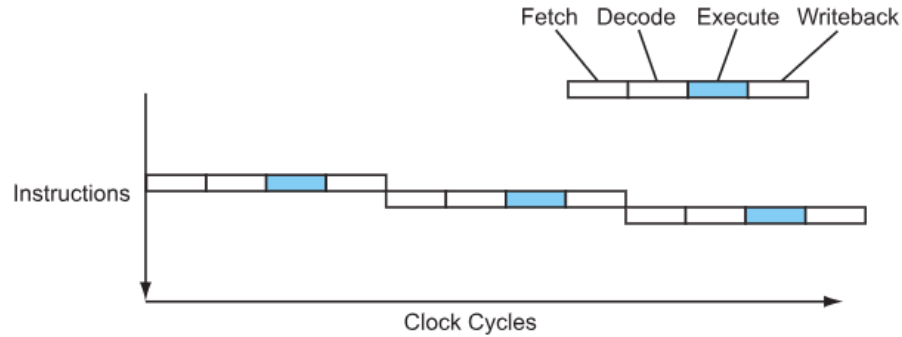


Figure 1 – The instruction flow of a sequential processor.

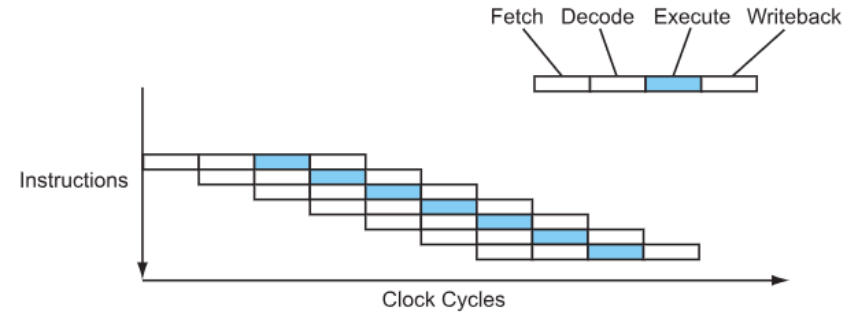


Figure 2 – The instruction flow of a pipelined processor.

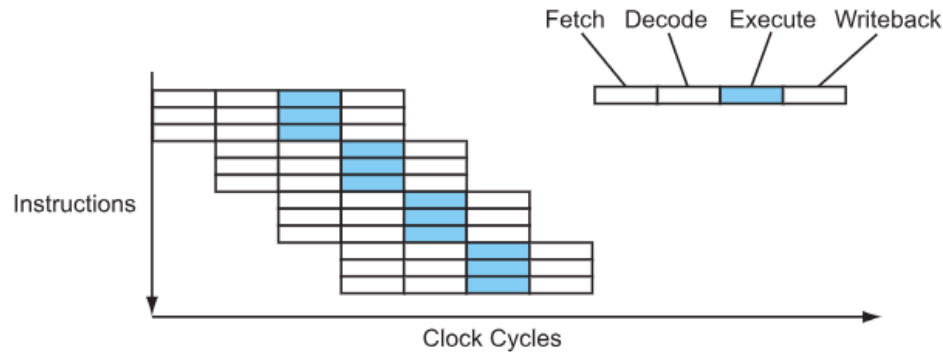
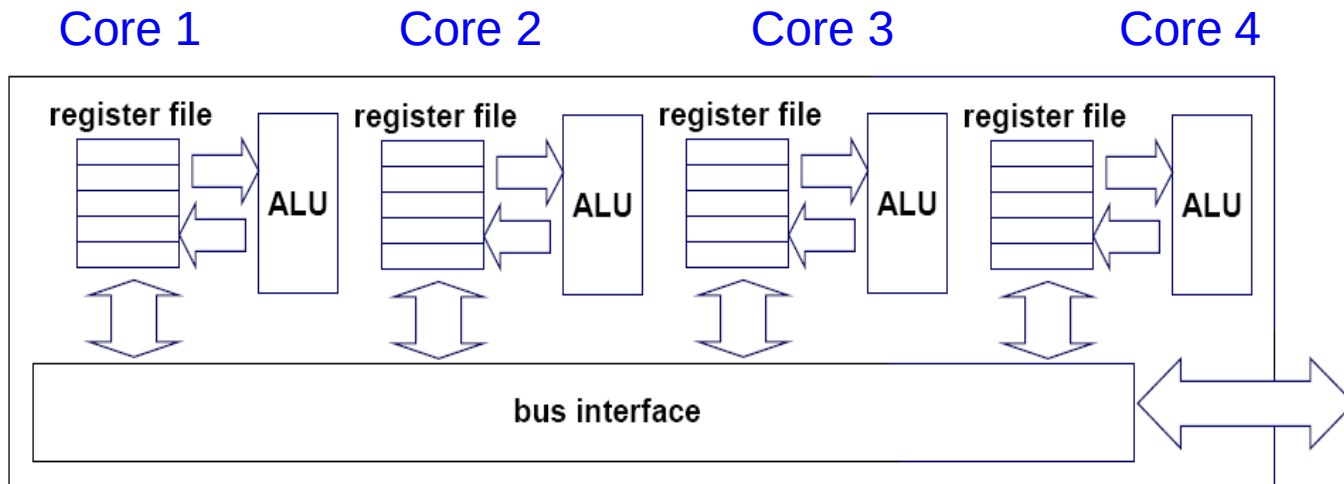
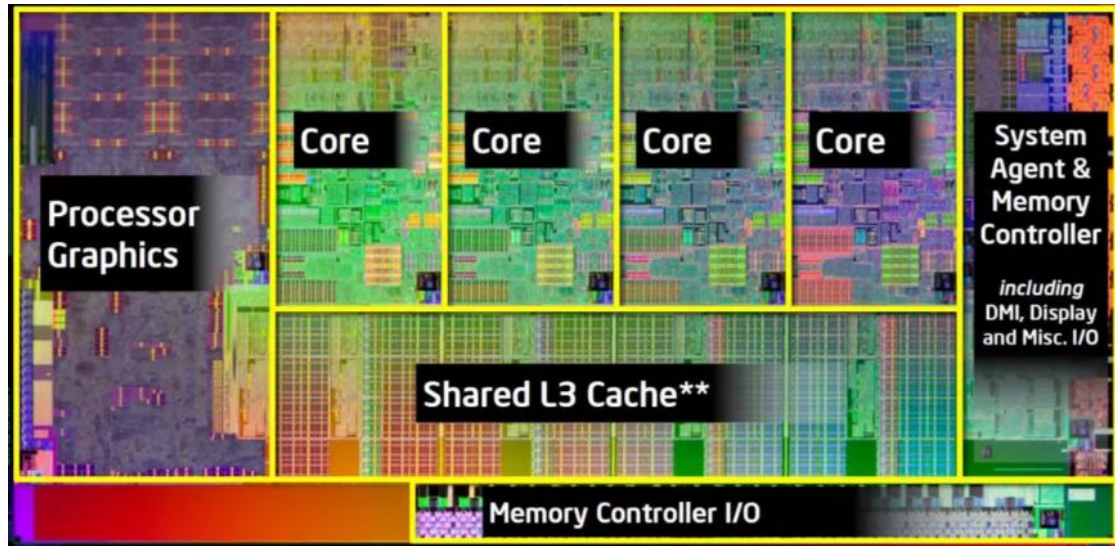


Figure 8 – The instruction flow of a superscalar processor.

Multicore Machines

- Core is a unit that reads and executes program instructions
- Processors were originally developed with only one core
- A multi-core processor is a single computing component with two or more independent processors ("cores").
- Replicate multiple processor cores on a single die.
- Manufacturers typically integrate cores onto a single integrated circuit die (chip multiprocessor or CMP), or onto multiple dies in a single chip package.







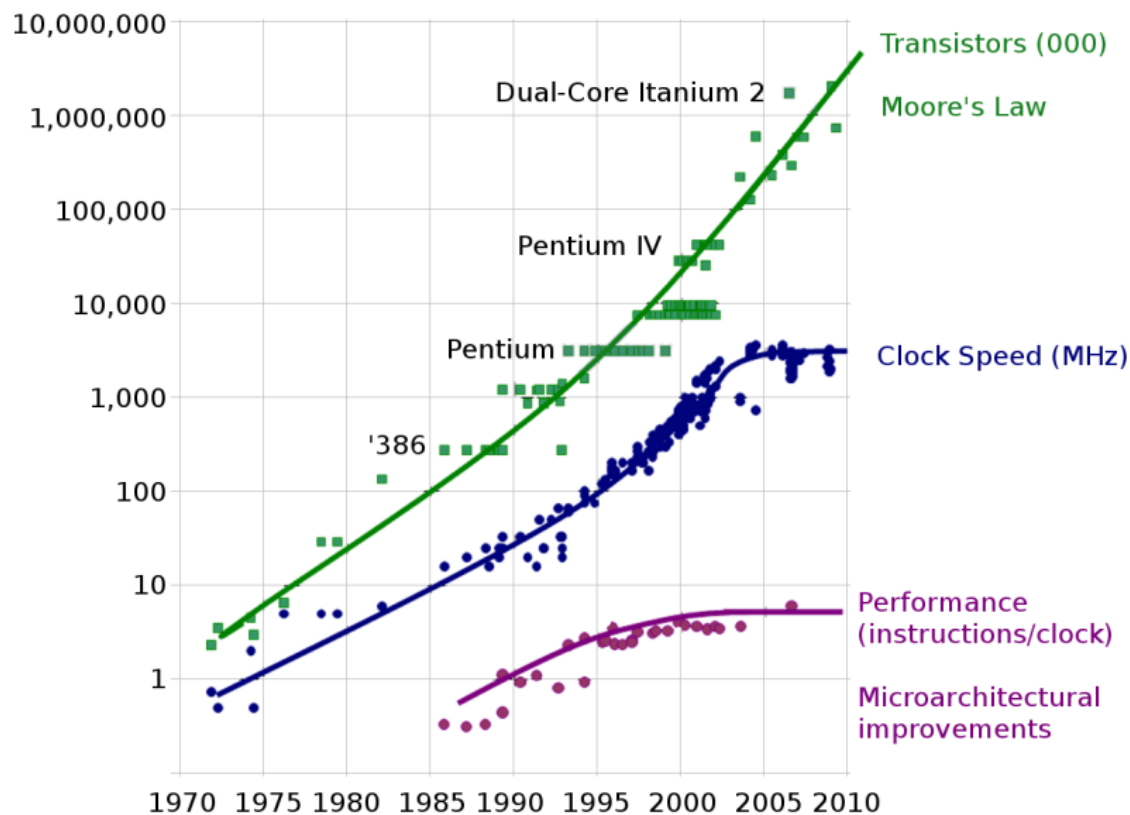
Why Multicores?



During 50 yrs of Moore's Law: transistor density X 2 every 2 years

- Transistors grew smaller
- Smaller transistors require less voltage and current
 - Can switch at higher frequency at same power
 - Clock rates increased
 - Capabilities could be added/expanded at constant power
 - Instruction-level parallelism
 - OO executions
 - Prefetching
 - More cache
- Single processor performance X 2 every 18 months

Intel CPU Trends



Sutter, *The Free Lunch is Over*, DDJ 2005.

Data: Intel, Wikipedia, K. Olukotun



Why Multicores?



Limits to single processor performance

- Moore's Law- CPU doubles its frequency every 2 yrs.
 - Possible by increasing the transistor density
 - Difficult to sustain because of heat produced
- Power increase with transistor size
- Power per transistor cannot be reduced
 - More transistors packed in same area -> higher power density -> more heat
 - Faster processors get too hot(~171F at 3.8GHz), cooling gets bulky and expensive
 - Speed has not really increased (~ 4 GHz max)
 - Adding lower frequency cores keeps computational capacity increasing at comparable power



Why Multicores?

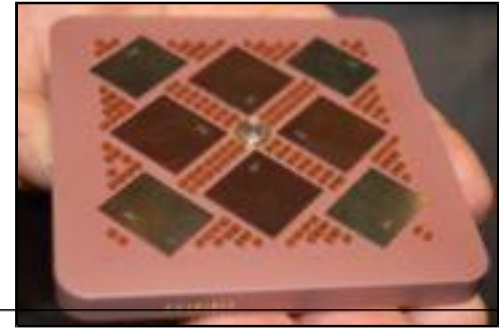


Logical extension is Multicore

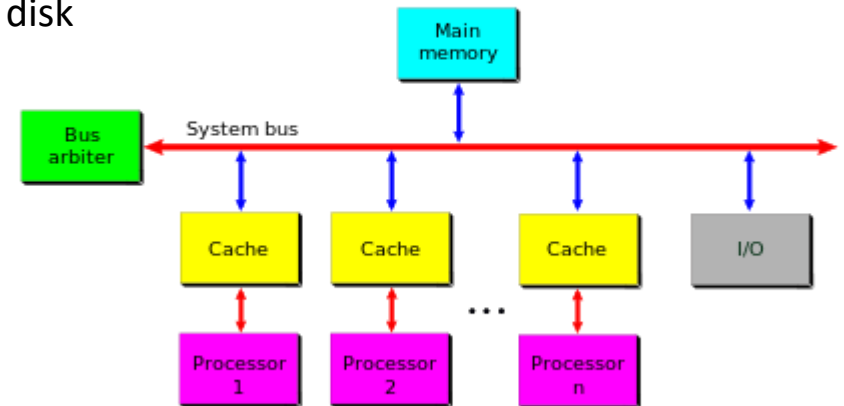
- 100s relatively slow(1.5GHz) processors vs 10s relatively fast(2.5GHz) processors
- Overall more compute available for comparable power
- Multicore is more general purpose
- A multi-core processor is an integrated circuit to which two or more processors have been attached for enhanced performance, reduced power consumption and more efficient simultaneous processing of multiple tasks.

Symmetric Multiprocessors (SMP)

- Two or more identical/ homogenous processors are connected to a single shared main memory
- Inter-processors communication takes place through shared memory
- Shared memory hardware architecture where multiple processors share a single address space and have equal access to all resources - memory, disk

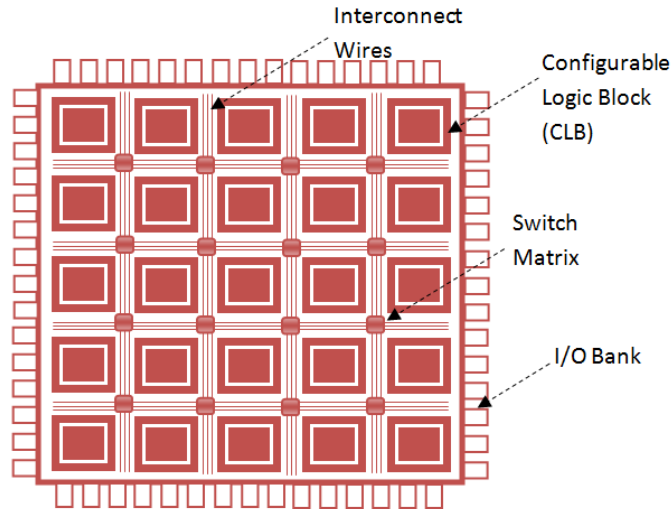


SMP - symmetric multiprocessor system



Field-Programmable Gate Arrays

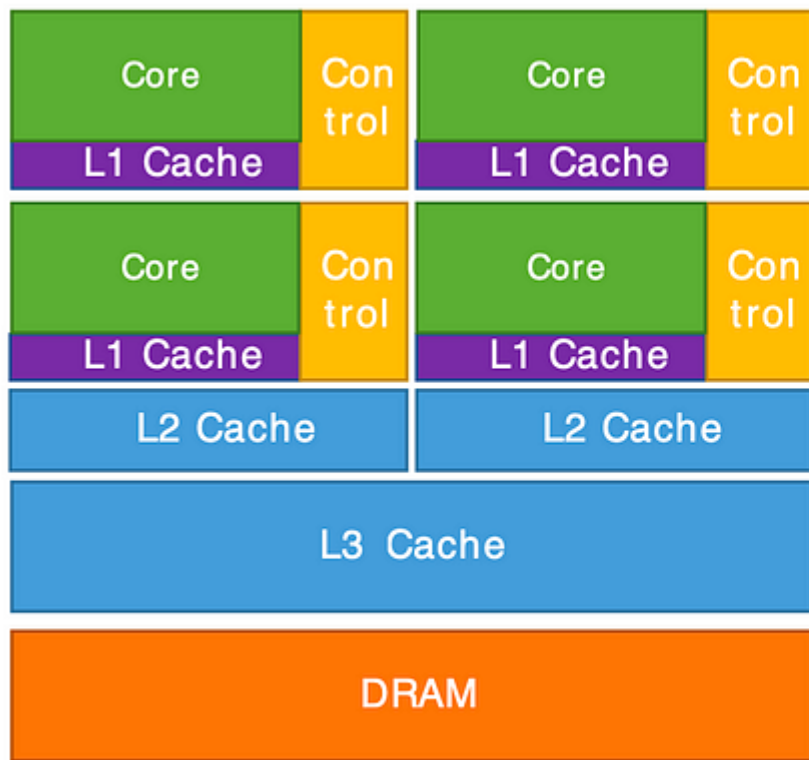
- a computer chip that can rewire itself for a given task
- can be programmed with hardware description languages such as VHDL or Verilog



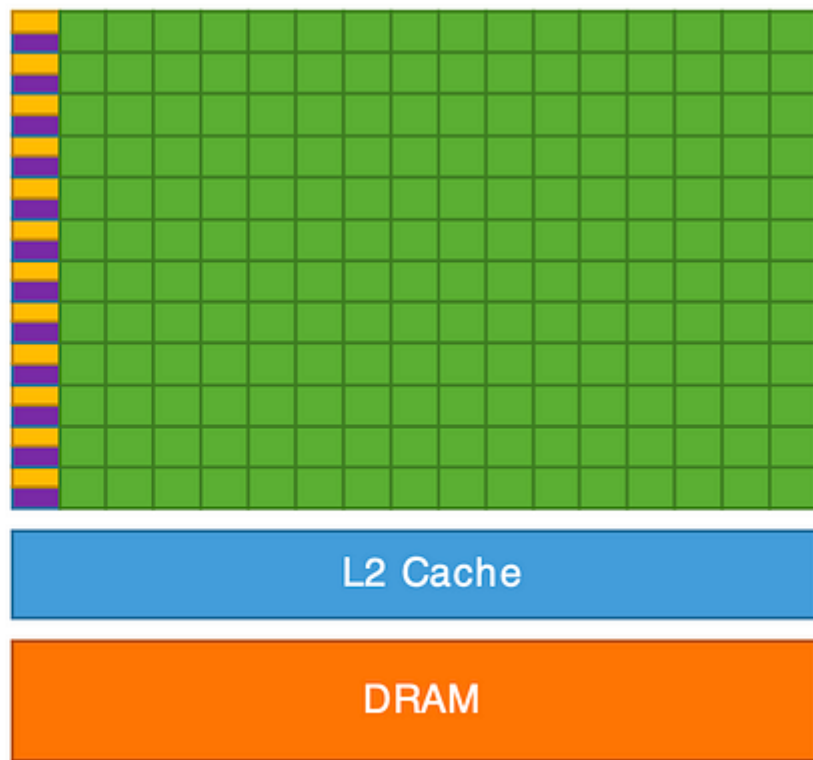
General Purpose GPU

- General-purpose computing on graphics processing units (GPGPU)
- NVIDIA, AMD and Intel
- Nvidia GPU cores are known as streaming multiprocessors.
- CUDA/OpenCL programming environment





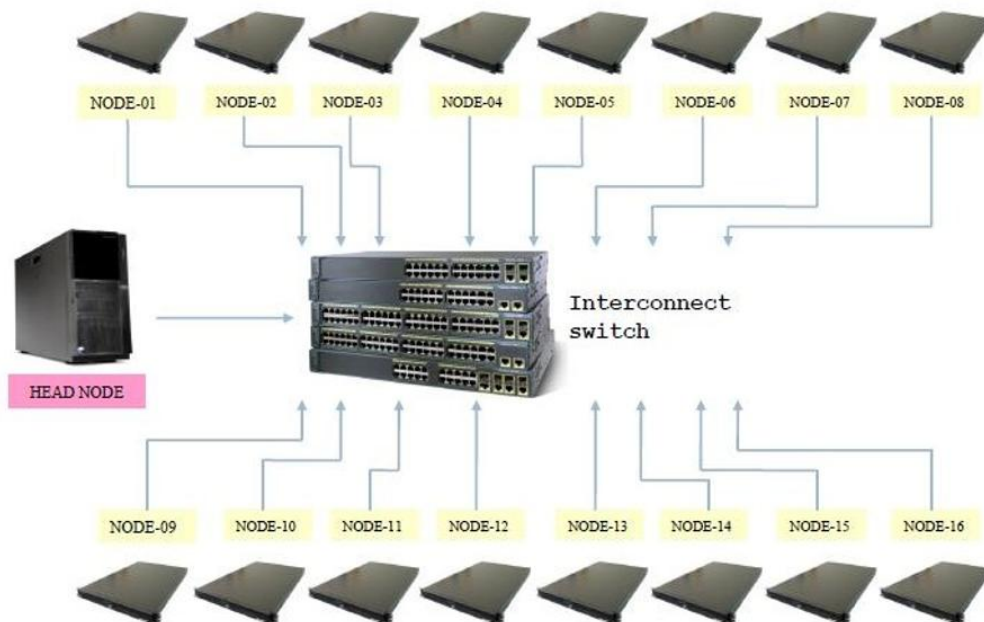
CPU



GPU

- Tightly coupled computers that work together closely as though they are a single computer.

Clusters



Supercomputer Clusters

- Tightly coupled computers that work together closely as though they are a single computer.





Cluster Computing



Processing

- In general, there are two parallel processing approaches: symmetric multiprocessing (SMP) and massively parallel processing (MPP).

Interconnects

- In order for a large number of processors to work together, supercomputers utilize specialized network interfaces. These interconnects support high bandwidth and very low latency communication.
- Interconnects join nodes inside the supercomputer together.
- Popular Interconnects include GigE & Infiniband

Operating Systems

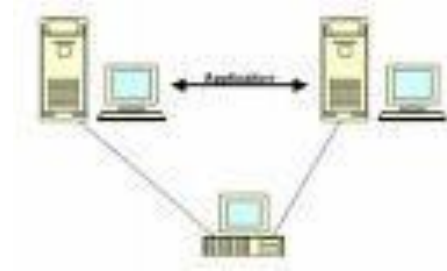
- Supercomputers today most often use variants of the Linux operating system.
- "lightweight" operating systems that consist of a small, simple kernel with many of the capabilities of a general-purpose O/S removed.

Programming

- The parallel architectures of supercomputers demand use of special programming techniques to exploit their speed.
- The base language of supercomputer code is, in general, Fortran or C, using special libraries to share data between nodes.
- Environments such as MPI and PVM for loosely connected clusters and OpenMP for tightly coordinated shared memory machines are used.
- GPGPUs have hundreds of processor cores and are programmed using programming models such as CUDA and OpenCL.

Distributed Computing

- different parts of a program are run simultaneously on separate computers communicating over a network



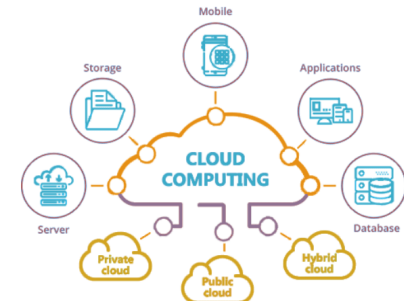
Grid Computing

- Aggregation, sharing of distributed heterogeneous systems



Cloud computing

- on-demand available service- IaaS, PaaS, SaaS
- pay-as-you-use over the internet



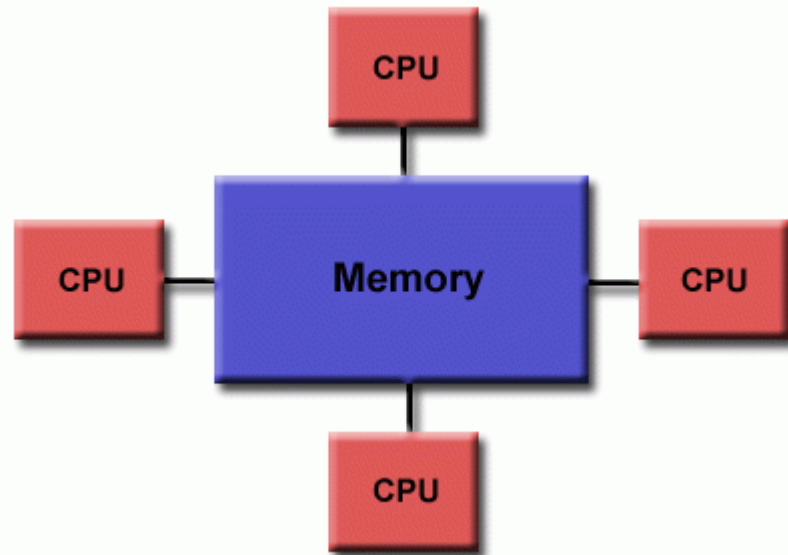


Memory Architectures

Shared Memory Architecture

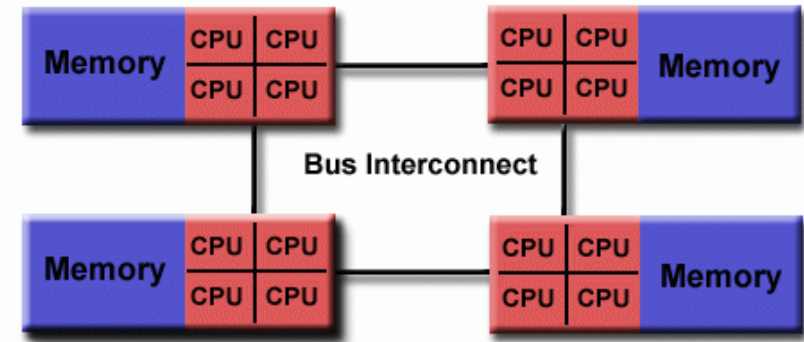
Uniform Memory Access (UMA):

- Common example - Symmetric Multiprocessor (SMP)
- Identical processors
- Equal access and access times to memory
- Sometimes called CC-UMA - Cache Coherent UMA. Cache coherent means if one processor updates a location in shared memory, all the other processors know about the update. Cache coherency is accomplished at the hardware level.
- Contention - as more CPUs are added, competition for access to the bus leads to a decline in performance.
- Thus, scalability ~32 processors



Non-Uniform Memory Access (NUMA):

- Not all processors have equal access time to all memories
- Memory access across link is slower
- If cache coherency is maintained, then called CC-NUMA
- Designed to overcome scalability limits of SMPs.
- Can support upto 1024 processors
- Processors directly attached to a memory module experience lower latency than those attached to “remote” memory modules



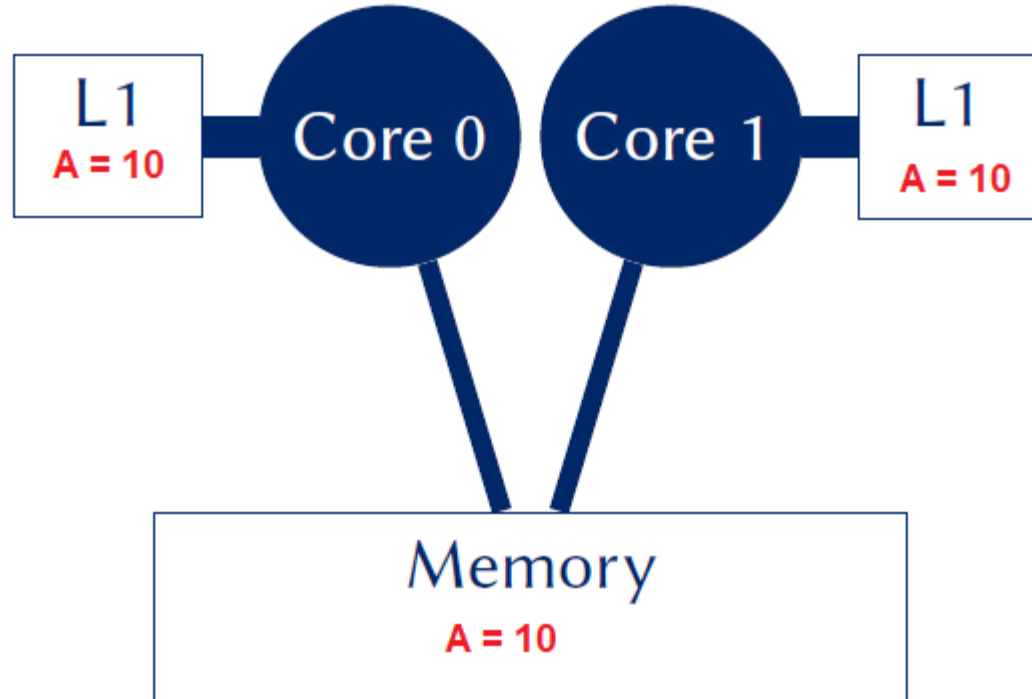


Cache Coherence in Multiprocessors

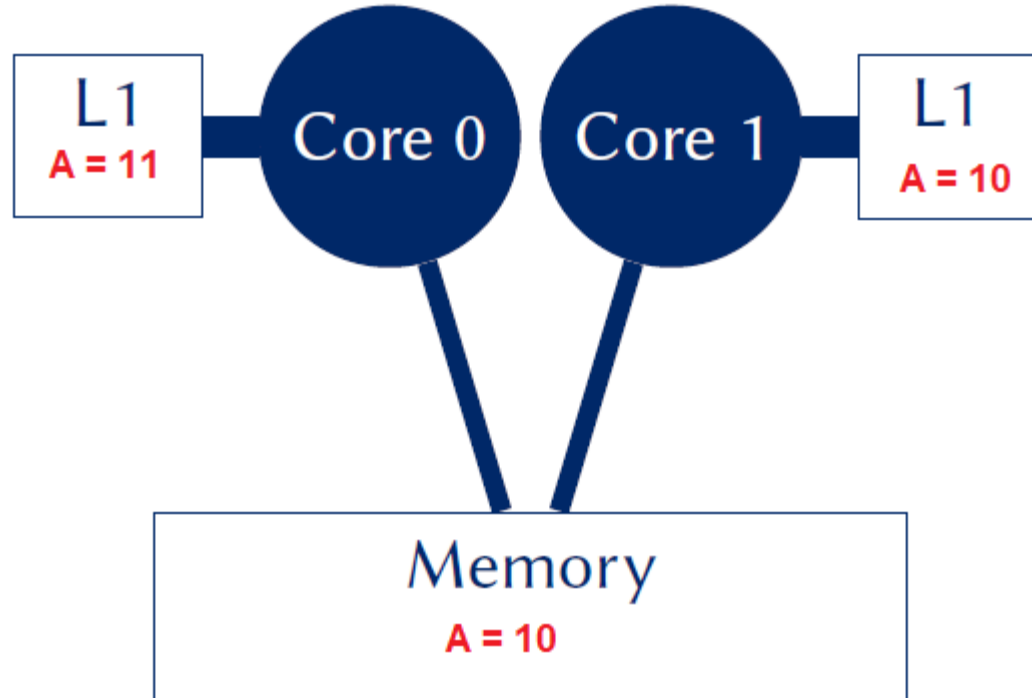


- Maintaining coherence between data loaded in processor caches is an issue in multiprocessor system
 - Same memory block is loaded into two processor cache
 - One of the processor updates the data in its local cache
 - Data in the other processor cache and in memory becomes inconsistent

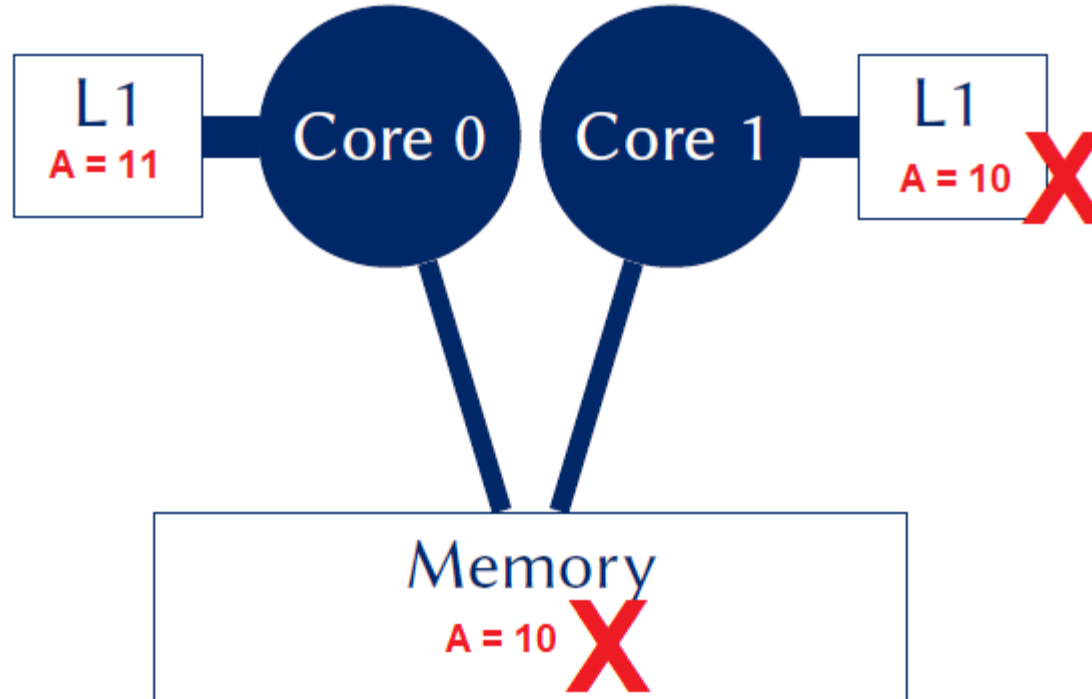
Cache Coherence



Cache Coherence

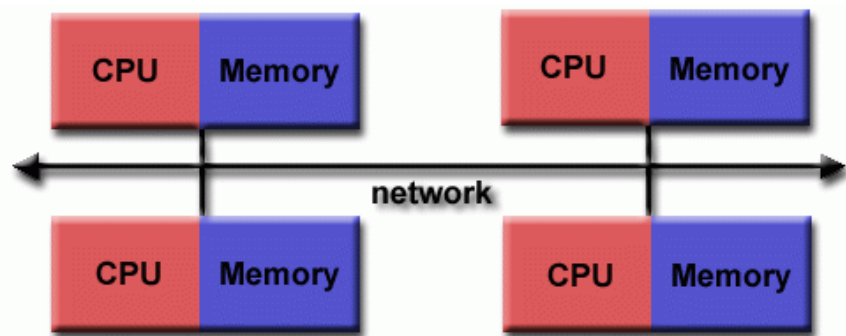


Cache Coherence



Distributed Memory Architecture

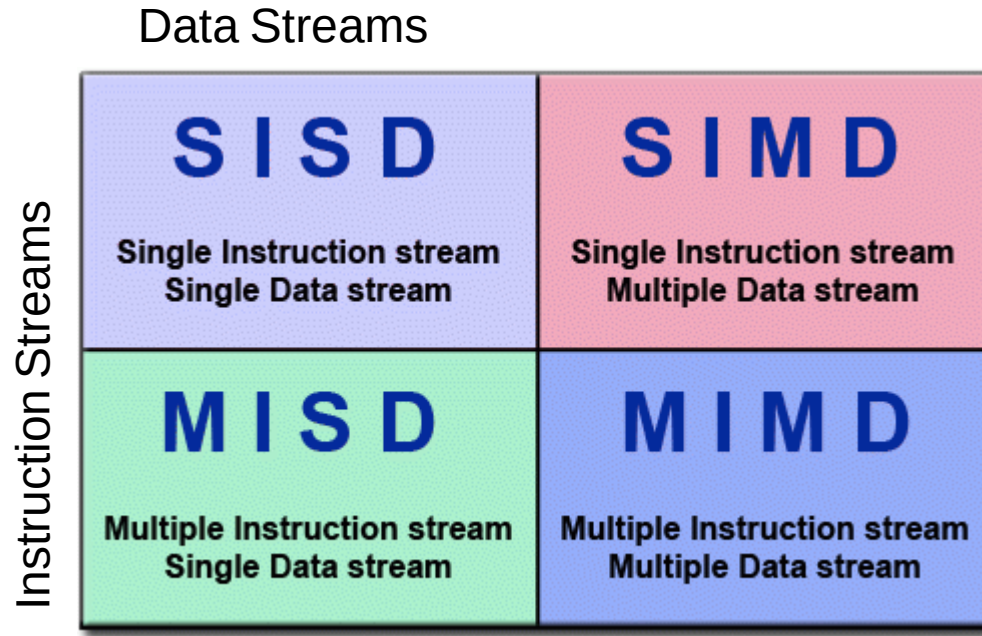
- Processors have their own local memory. No concept of global address space across all processors.
- Because each processor has its own local memory, it operates independently. Changes it makes to its local memory have no effect on the memory of other processors. Hence, the concept of cache coherency does not apply.
- When a processor needs access to data in another processor, it is usually the task of the programmer to explicitly define how and when data is communicated. Synchronization between tasks is likewise the programmer's responsibility.
- The network "fabric" used for data transfer varies widely, though it can be as simple as Ethernet.





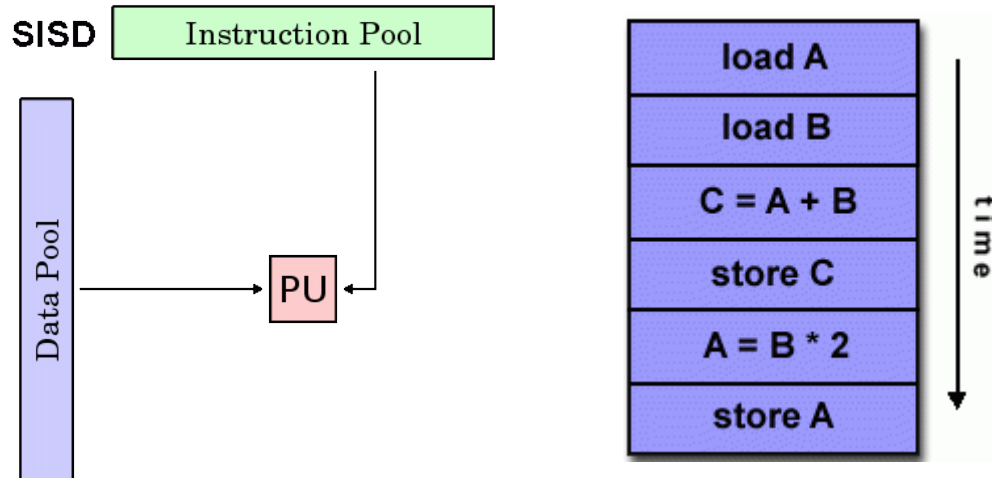
Parallel Computing Classifications

Flynn's Taxonomy



Single Instruction Single Data

- One processing element
- Single program and single data source
- Simple processor with no parallelism



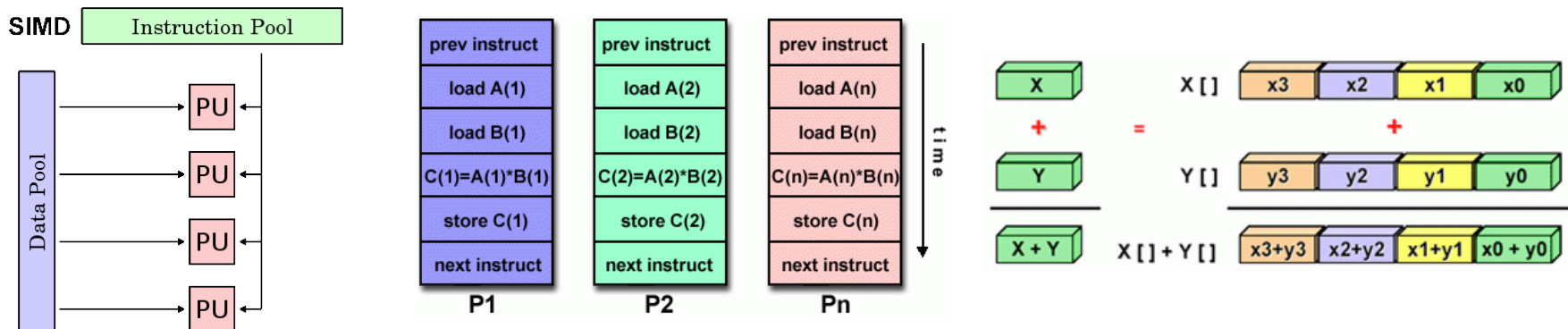
UNIVAC



IBM 360

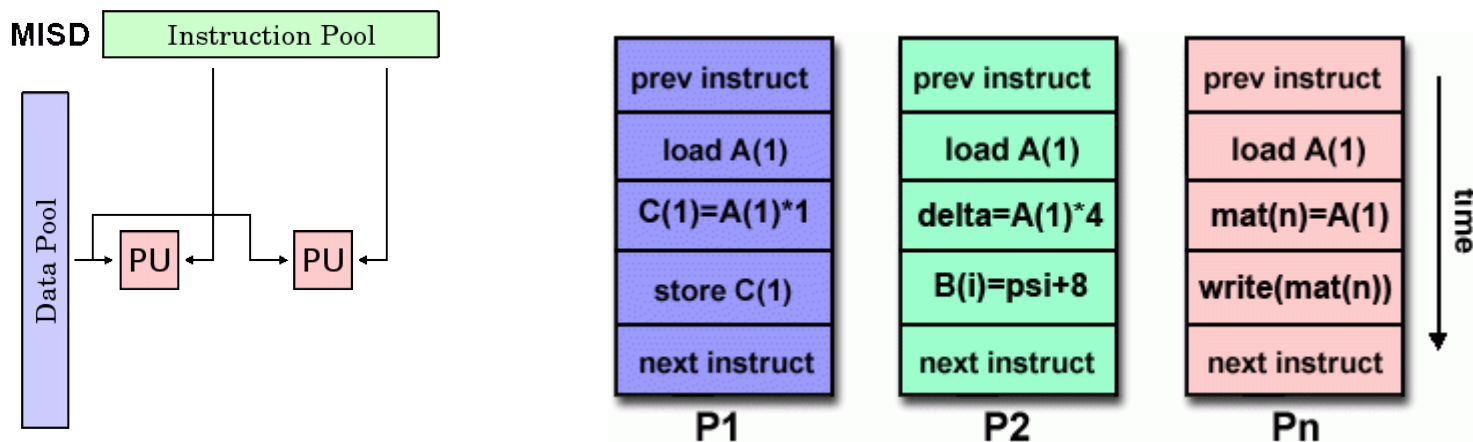
Single Instruction Multiple Data

- Multiple processing element or unit
- Each one executes on its data. The physical memory might be shared
- All execute same instructions at the same time. Operations on vectors
- Basic architecture of GPUs
- specialized problems characterized by a high degree of regularity such as graphics/image processing
- Vector Pipelines: IBM 9000, Cray X-MP, Y-MP & C90, Fujitsu VP, NEC SX-2, Hitachi S820, ETA10
- Most modern computers, particularly those with graphics processor units (GPUs) employ SIMD instructions and execution units.



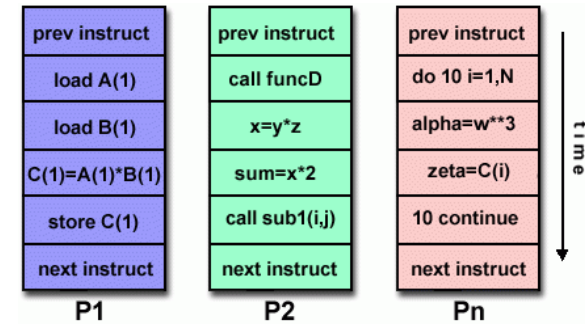
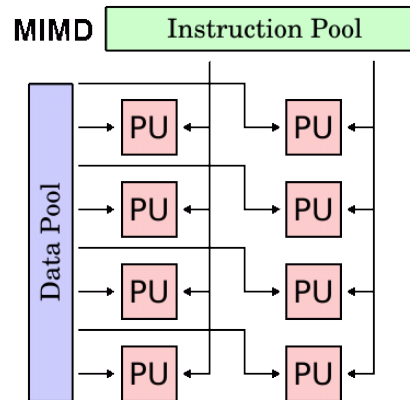
Multiple Instruction Single Data

- Multiple processing element or unit
- Each one executes a different program on the same data
- Not a very common case
- multiple cryptography algorithms attempting to crack a single coded message
- multiple frequency filters operating on a single signal stream

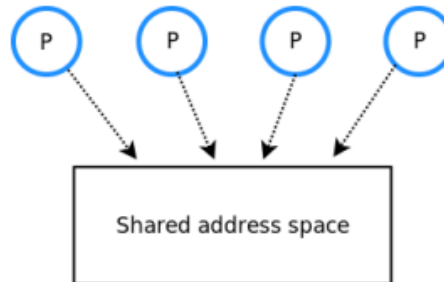




- A type of parallel computer
- Every processor may be executing a different instruction stream
- Every processor may be working with a different data stream
- Execution can be synchronous or asynchronous, deterministic or non-deterministic
- Supercomputers fall into this category, networked parallel computer clusters and grids, multi-processor SMP computers, multi-core PCs.
- Many MIMD architectures also include SIMD execution sub-components

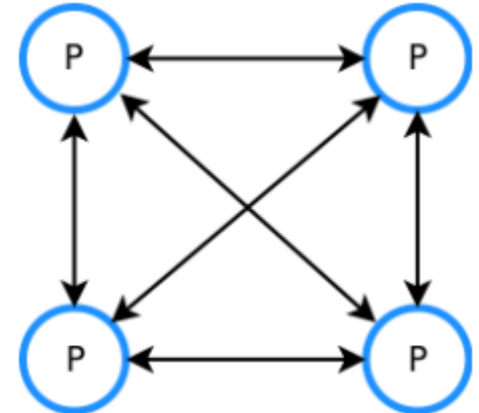


- A multicore processor with coherent shared memory
- Read and write operations in the shared address space
- Synchronization primitives (locks, semaphores etc)
- Simplest way of parallelizing a sequential program
 - All communications are implicit
 - Achieving good performance can be challenging
- Threads / processes communicate by running operations in shared address space



Message passing

- Threads/processes communicate by sending and receiving messages
- All communications are explicit
 - More effort for the programmer
 - Allow to better optimize performance
 - Control on the data movements
 - Better overlap between communication and computation
- A set of computers interconnected through a network
- Also called distributed memory system



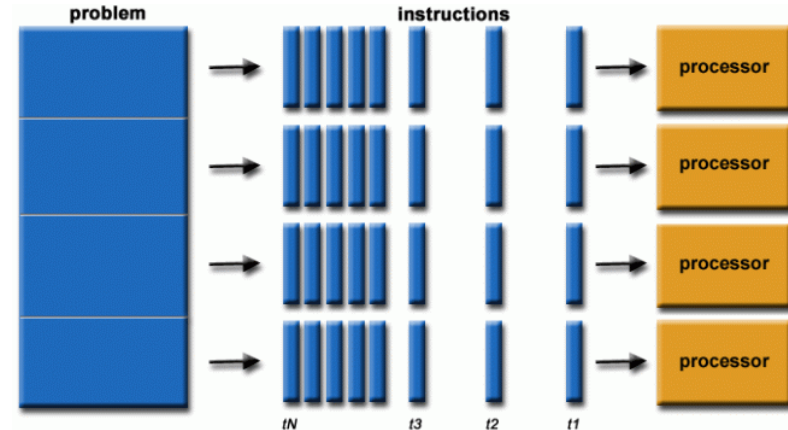
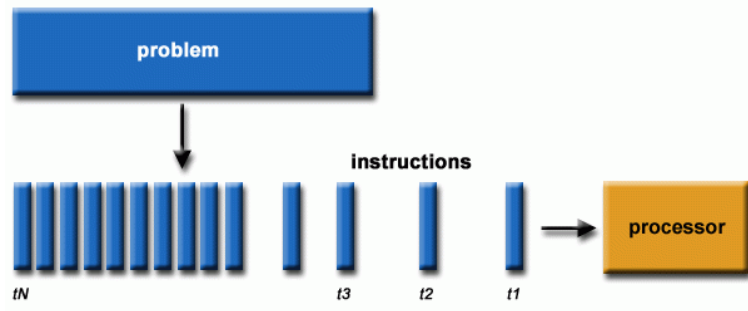


Parallel Programming Models



- On a single machine with shared memory
 - Pthreads
 - OpenMP
- On machines connected via network and have no shared memory
 - MPI
 - Partitioned Global Address Space (PGAS)
- Accelerators, offload tasks from CPU to it
 - CUDA
 - OpenACC
 - OpenCL
- Hybrid
 - OpenMP + MPI
 - CUDA + MPI

Parallel Programming



- A parallel processing program is a single program that runs on multiple processors simultaneously.
- The overall problem is split into parts, each of which is performed by a separate processor in parallel.
- In addition to faster solution, it may also generate a more precise solution



Terminologies



- Parallel Processing
 - Processing multiple tasks simultaneously on multiple processors is called parallel processing
- Parallel Programming
 - Software methodology used to implement parallel processing
- Parallel Computing
 - Encompasses all the technologies used in running multiple tasks simultaneously on multiple processors



High Performance Computing

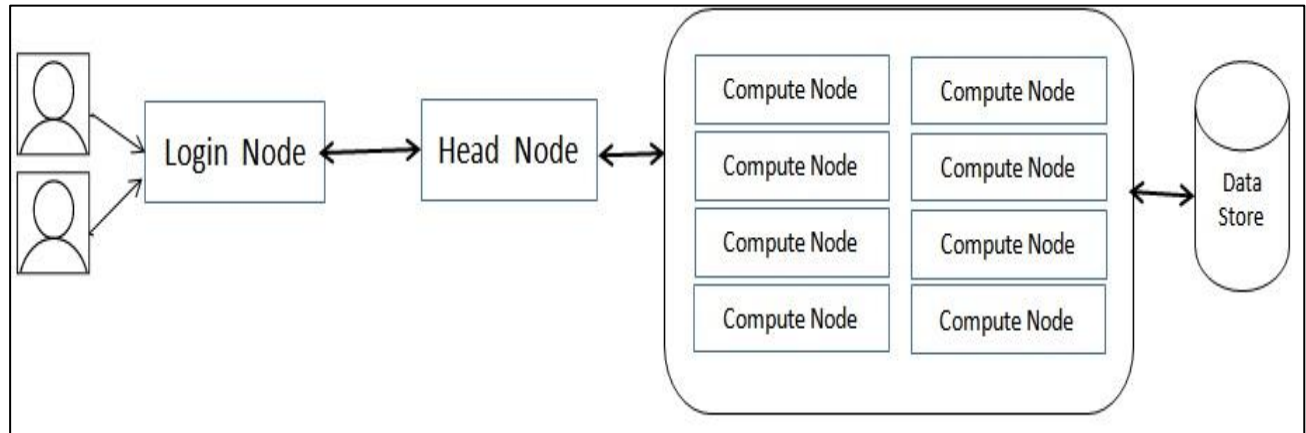


- High performance computing (HPC) is the ability to process data and perform complex calculations at high speeds
- HPC refers to high-speed parallel processing of complex computations on multiple servers. A group of these servers is called a cluster, which consists of hundreds or thousands of computing servers connected through a network
- HPC is a computer with a high-level computational capacity compared to a general purpose computer. Aggregation of computing power Delivers much higher performance than one could get out of a regular desktop

HPC components

Three main components

- Compute
- Network
- Storage





Units of HPC



Weather Forecasting

If 1 cell is of size 1km x 1km x 1km

The whole atmosphere about 5×10^8 cells

- 1 Mflop/s 10^6
- 1 Gflop/s 10^9
- 1 Tflop/s 10^{12}
- 1 Pflop/s 10^{15}
- 1 Eflop/s 10^{18}
- 1 Zflop/s 10^{21}
- 1 Yflop/s 10^{24}

If each calculations require 200 Flops,

- it will take 200 FLOPS/cell * 5×10^8 cells for each time step
- Number of time steps for a 10-day simulation with 10-minute intervals (144 intervals/day) will be 1440
- Total FLOPS = (Number of FLOPS per time step per cell) * (Number of cells) * (Number of time steps)
- Total FLOPS = (200 FLOPS/cell) * (5×10^8 cells) * (1440 time steps)
- Total FLOPS = 1.44×10^{17} FLOPS
- **It would take approximately 1,666 days to compute the simulation with a 100 MFLOPS system and about 20 hours to compute the simulation with a 2 TFLOPS system.**

Performance in FLOPS with scale

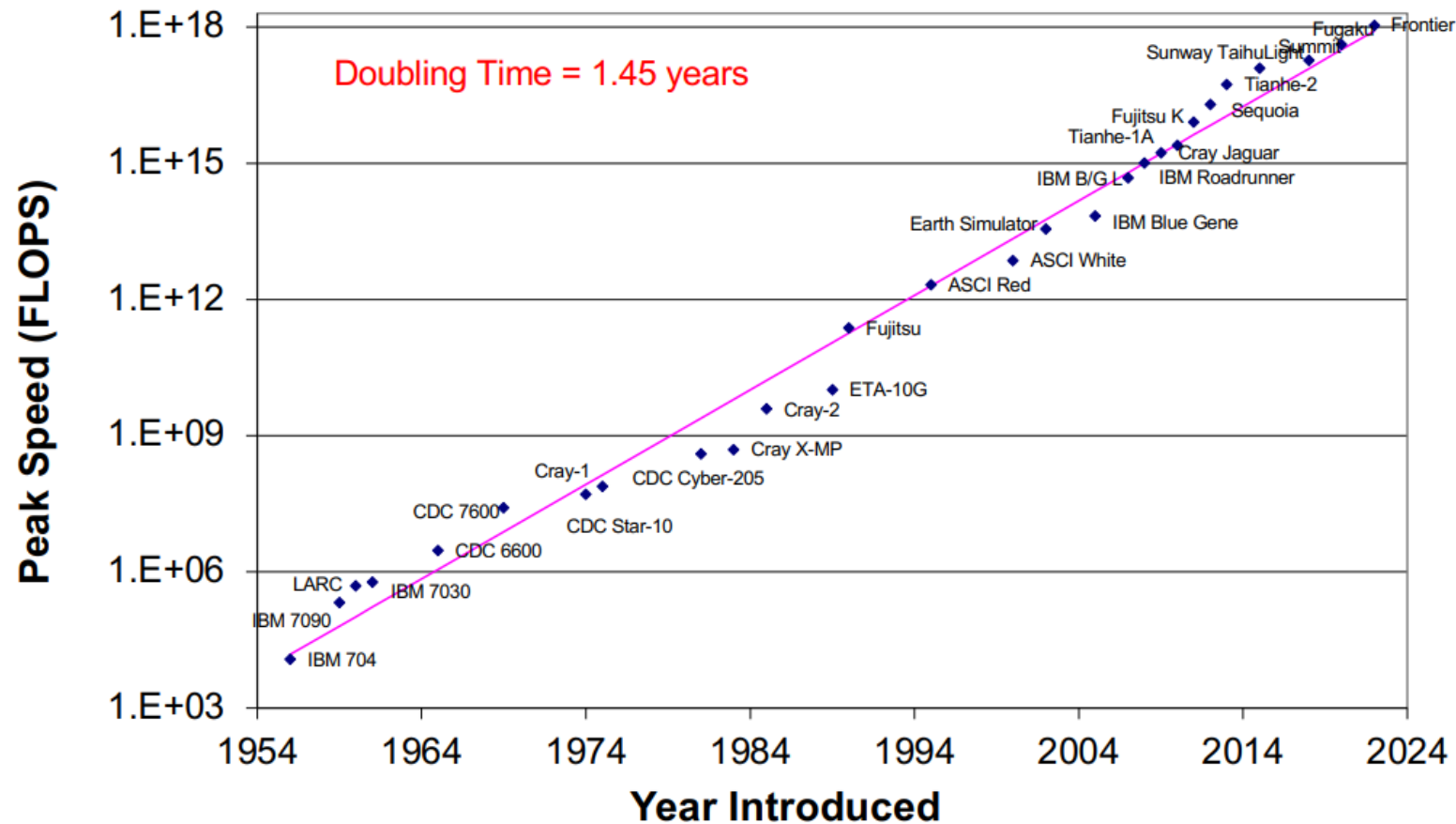
Name	Notation	Exponent	Example
kiloFlops	1,000	10^3	-
megaFlops	1,000,000	10^6	-
gigaFlops	1,000,000,000	10^9	Average Laptop range
teraFlops	1,000,000,000,000	10^{12}	Newest gaming console and graphic cards
petaFlops	1,000,000,000,000,000	10^{15}	Supercomputer range
exaFlops	1,000,000,000,000,000,000	10^{18}	Most Powerful Supercomputer
zettaFlops	1,000,000,000,000,000,000,000	10^{21}	-
yottaFlops	1,000,000,000,000,000,000,000,000	10^{24}	-

Supercomputer

- Supercomputers are a class of high-performance computing (HPC) systems designed to offer higher computational power and processing speed.
- Tightly coupled computers that work together closely as though they are a single computer



Growth of Supercomputer



World Top5 Supercomputers(<https://top500.org/>)

Rank	Site	System	Cores	Rmax (PFlop)	RPeak (PFlop)
1	Frontier , DOE/SC/Oak Ridge National Laboratory United States	HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE	8,699,904	1,194.00	1,679.82
2	Aurora, DOE/SC/Argonne National Laboratory United States	HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel	4,742,808	585.34	1,059.33
3	Eagle, Microsoft Azure United States	Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR	1,123,200	561.20	846.84
4.	A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu	A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu	7,630,848	442.01	537.21
5.	LUMI, EuroHPC/CSC Finland	HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE	2,220,288	309.10	428.70



Terminology



- **FLOPS (Floating Point Operations Per Second)**: FLOPS is a fundamental measure of computational speed, representing the number of floating-point calculations a supercomputer can perform in one second. It is often used to quantify a supercomputer's raw processing power.
- **Top500 List**: The Top500 list is a semiannual ranking of the world's most powerful supercomputers. It considers the LINPACK benchmark results and ranks supercomputers based on their performance. The top-ranked supercomputer is considered the fastest in the world.
- **Rmax**: metric that measures the maximum sustained performance of a supercomputer when running real-world applications and simulations. It is typically expressed in FLOPS.
- **Rpeak**: represents the theoretical peak performance of a supercomputer, indicating the maximum computational speed it can achieve in ideal conditions. This is usually higher than the real-world Rmax performance.



Terminology

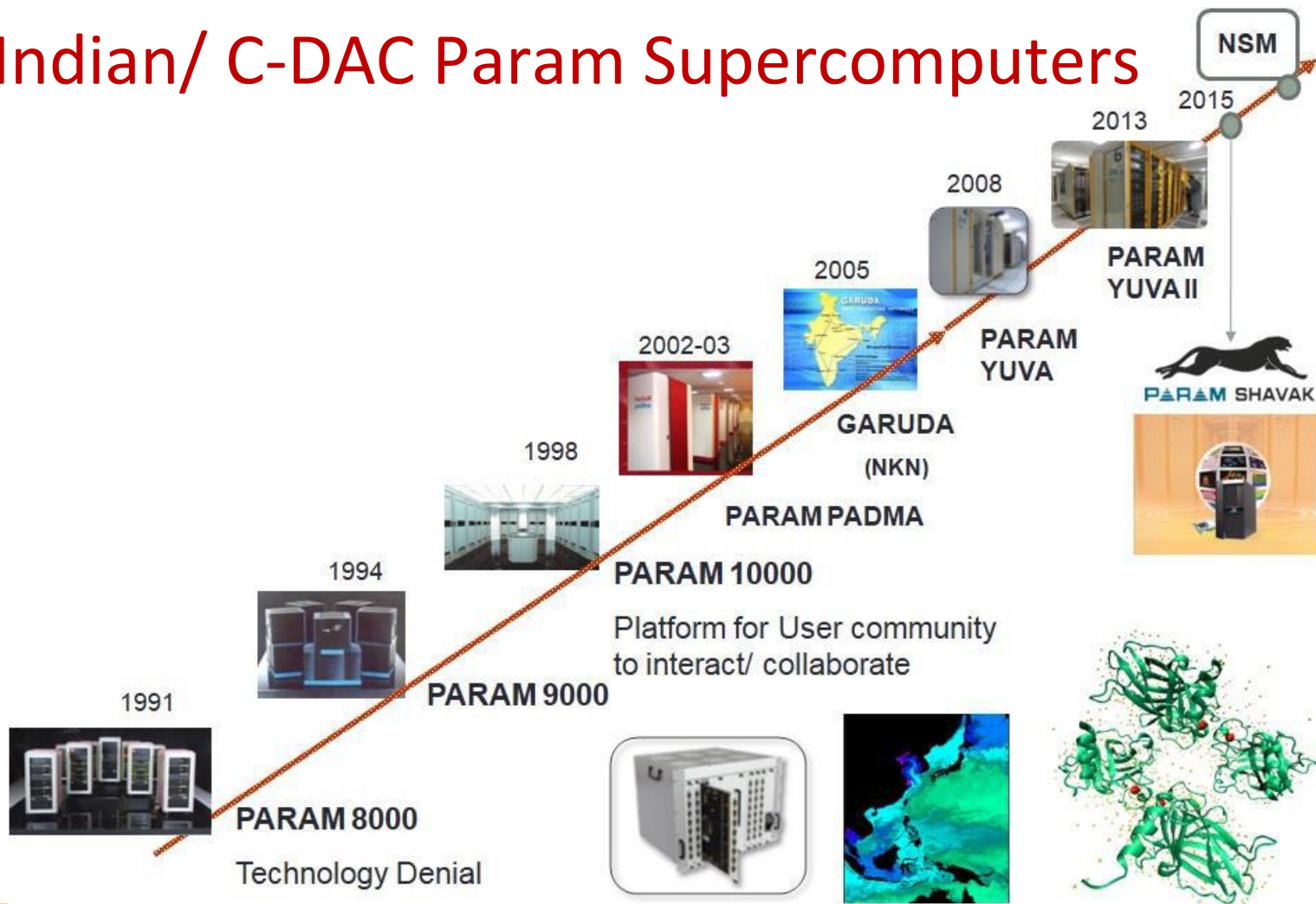


- **LINPACK Benchmark:** The LINPACK benchmark is a widely used method to assess a supercomputer's performance. It measures how quickly a system can solve a dense system of linear equations and provides a standardized way to compare different supercomputers.
- **HPL (High-Performance Linpack):** HPL is a variant of the LINPACK benchmark used in the Top500 list. It is a standardized test to evaluate a supercomputer's performance.
- **Energy Efficiency:** Energy efficiency measures how efficiently a supercomputer uses power to perform calculations. It is expressed as FLOPS per watt and helps determine the environmental impact and cost-effectiveness of a supercomputer.
- **Scalability:** Scalability assesses a supercomputer's ability to handle larger workloads by adding more processing nodes or components. Scalability is crucial for accommodating growing computational demands.

Year	Name	Peak speed	Location
1960	LARC	500 kFLOPS (2 CPUs)	Lawrence Livermore Lab. USA
1961	IBM 7030	1.2 MIPS 600 kFLOPS	Los Alamos Lab. USA
1965	CDC 6600	10 MIPS 3 MFLOPS (10^6)	Lawrence Livermore Lab. USA
1965	CDC 7600	36 MFLOPS	Lawrence Livermore Lab. USA
1975	Cray-1	80 MFLOPS (vector) 72 MFLOPS (scalar)	Los Alamos Lab. USA
1985	Cray-2	1.95 GFLOPS (4 CPUs) 3.9 GFLOPS (8 CPUs) (10^9)	Lawrence Livermore Lab. USA
1995	Intel ASCI Red	2.15 TFLOPS (10^{12})	Sandia National Lab. USA
2010	Tianhe-1A	2.57 PFLOPS (10^{15})	National Supercomputer Centre, Tianjin, China
2022	Frontier	1102 PFLOPS (10^{18})	Oak Ridge National Lab, USA



Indian/ C-DAC Param Supercomputers





Indian Top5 Supercomputers

(<https://topsc.cdacb.in>)

Rank	Site	System	Cores/ nodes	Rmax (TFlop)	RPeak (TFlop)
1 [90]	Airawat-PSAI, C-DAC Pune	NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHZ, NVIDIA A100, INFINIBAND HDR2	81344	8500	13176
2 [163]	PARAM Siddhi-AI, C-DAC,Pune	NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHZ, NVIDIA A100, Mellanox HDR Infiniband (OEM:ATOS under NSM initiative, Bidder)	41664/ 236	4619	5267.14
3 [201]	Pratyush, Indian Institute of Tropical Meteorology, Pune	Cray XC-40 class system with 3315 CPU-only (Intel Xeon Broadwell E5-2695 v4 CPU) nodes with Cray Linux environment as OS, and connected by Cray Aries interconnect	119232/ --	3763.9	4006.19
4. [354]	Mihir, NCMRWF, Noida	Intel Xeon Broadwell E5-2695 v4 CPU with Cray Linux environment connected by Cray Aries interconnect OEM:Cray, Bidder:Cray	83592/ 1152	2570.4	2808.7
5.	PARAM Pravega, IISc, Bangalore	Intel Xeon Cascade Lake processors,NVIDIA Tesla V100 with NVLink, Mellanox HDR interconnect. OEM:Atos, Bidder:Atos	29952/ 624	1702	2565



NATIONAL SUPERCOMPUTING MISSION

INFRASTRUCTURE | APPLICATIONS | R&D | HRD



- Jointly Steered by Department of Science and Technology (DST) and Ministry of Electronics and Information Technology (MeitY)

Mission Statement:

- The Mission envisages empowering our national academic and R&D institutions spread over the country by installing a vast supercomputing grid comprising of more than 70 high-performance computing facilities.

Objective:

- To make India one of the world leaders in Supercomputing and to enhance India's capability in solving grand challenge problems of national and global relevance
- To empower our scientists and researchers with state-of-the-art supercomputing facilities and enable them to carry out cutting-edge research in their respective domains
- To minimize redundancies and duplication of efforts, and optimize investments in supercomputing
- To attain global competitiveness and ensure self-reliance in the strategic area of supercomputing technology

NSM Sites

- PARAM Shivay, IIT BHU, 837 TFLOPS
- PARAM Shakti, IIT Kharagpur, 1.66 PFLOPS
- PARAM Brahma, IISER Pune, 797 TFLOPS
- PARAM Yukti, JNCASR Bengaluru, 838 TFLOPS
- PARAM Sanganak, IIT Kanpur, 1.67 PFLOPS
- PARAM Pravega, IISc Bangalore, 3.3 PFLOPS
- PARAM Seva, IIT Hyderabad, 838 TFLOPS
- PARAM Smriti, NABI, Mohali, 838 TFLOPS
- PARAM Utkarsh, C-DAC Bengaluru, 838 TFLOPS
- PARAM Siddhi-AI, C-DAC Pune,
- PARAM Ganga, IIT Roorkee, 1.67 PFLOPS
- PARAM Ananta, IIT Gandhinagar, 838 TFLOPS
- PARAM Porul, NIT Trichy, 838 TFLOPS
- PARAM Himalaya, IIT Mandi, 838 TFLOPS
- PARAM Kamrupa, IIT Guwahati, 838 TFLOPS

PARAM Shivay



PARAM Shakti



PARAM Brahma



PARAM Yukti



PARAM Sanganak



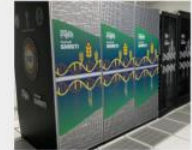
PARAM Pravega



PARAM Seva



PARAM Smriti



PARAM Utkarsh



PARAM Ganga



PARAM Ananta



PARAM Porul



PARAM Himalaya



PARAM Kamrupa





Param Utkarsh

परम
उत्कर्ष
PARAM
Utkarsh



Parallel Programming Concepts